Mathematical Modeling of Rotating Shaft Misalignment

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Scope and Simplifying Assumptions

In UAV propulsion systems, direct-drive brushless motors lack the flexible couplings found in industrial machinery. Therefore, misalignment is strictly scoped in this model as a rotating bent shaft (angular misalignment), where the rotor hub rotates at a permanent angular offset α relative to the true axis of rotation.

To provide a tractable simulation environment, the mathematical model is governed by the following explicit assumptions:

1. Exactly one motor is faulty; the other three are healthy.
2. All motors run at the same constant motor RPM (Ω) within each analysis window (e.g., steady hover).
3. The chassis is structurally symmetric, utilizing one impulse response per axis, identical for every arm, with $h_x(t) = h_y(t)$.
4. Healthy motors have zero residual mass unbalance, providing an optimistic theoretical lower bound for detection limits.
5. Parallel misalignment is omitted.

The 1Ω Radial Forcing Function (Dynamic Imbalance)

A bent shaft physically tilts the rotor. Because the rotor's center of gravity remains on the bent rotational axis, there is no simple static centrifugal unbalance. Instead, the tilted rotor generates a dynamic couple imbalance. The principal axis of inertia is no longer aligned with the spin axis, creating a rotating bending moment M_c. Given the rotor's polar inertia I_p and transverse inertia I_t, this moment is defined as:

$$M_c \approx (I_p - I_t) \sin\alpha \cos\alpha \cdot \Omega^2 \tag{1}$$

This moment must be reacted by the internal motor bearings. If the distance between the upper and lower bearings is L_b, the radial reaction force at the bearings is M_c/L_b.

Additionally, the tilted propeller disk redirects a component of its aerodynamic thrust T laterally. Because thrust scales with the square of the angular velocity ($T \propto \Omega^2$), this lateral aerodynamic force is approximately $T \sin \alpha$.

Both the couple imbalance and the aerodynamic thrust offset rotate at the fundamental running speed ($\Omega = 2\pi f_0$) and scale with Ω^2. Summing these contributions yields the total radial forcing amplitude F_1:

$$F_1 = \frac{(I_p - I_t) \sin \alpha \cos \alpha \cdot \Omega^2}{L_b} + T \sin \alpha \tag{2}$$

For a standard lightweight quadrotor, the inertia difference $(I_p - I_t)$ is tiny, meaning the aerodynamic thrust term $(T \sin \alpha)$ will dominate the F_1 radial response. Note that for counter-rotating motors on opposing arms, the phase sense of this rotating vector is inverted.

The 2Ω Axial Forcing Function (Phenomenological Model)

In classical machinery diagnostics, a 2Ω vibration signature is heavily correlated with misalignment. In direct-drive systems without couplings, this harmonic arises from anisotropic shaft stiffness (unequal bending stiffness in two perpendicular directions) and variations in axial bearing preload as the bent shaft acts as a cam during rotation.

This axial forcing is an assumed coupling into the Z-axis, not derived from first rigid-body principles. Rather than attempting a pure kinematic derivation, the axial force amplitude F_2 is treated as a phenomenological component. To satisfy dimensional consistency, it is modeled proportionally to a lateral stiffness anisotropy Δk_{lat} (in N/m), an effective lever arm L_{eff} (in meters), and the bend angle α:

$$F_2 = \Delta k_{lat} \cdot L_{eff} \cdot \alpha \tag{3}$$

System Integration

Because all motors are assumed to run at an identical, constant motor RPM (Ω), the structural sum collapses. A single accelerometer sees one overlapping set of peaks rather than four distinct sets. The continuous forcing functions for the faulty motor are defined as:

$$F_x(t) = F_1 \sin(\Omega t + \phi) \tag{4}$$

$$F_y(t) = F_1 \cos(\Omega t + \phi) \tag{5}$$

$$F_z(t) = F_2 \sin(2\Omega t + \phi_2) \tag{6}$$

When convolved with the system's symmetric structural impulse response $h(t)$ and combined with the baseline rigid-body acceleration and sensor noise $\eta(t)$, this generates the complex, multi-harmonic vibration environment represented in the simulated accelerometer signal.

Multi-Feature Fault Detection Logic

Because the radial force is a rotating vector and the chassis is assumed symmetric ($h_x = h_y$), shifting the fault to a different arm only alters the phase, not the spectral magnitude. Consequently, amplitude cannot localize the faulty motor, strictly limiting the scope of the algorithm to fault detection and type classification.

To robustly classify a bent shaft, the signal processing pipeline utilizes multi-feature fusion:

- **Order Tracking:** The frequency spectrum is normalized by the constant motor RPM Ω (sourced via ESC telemetry). While not needed to separate identically-spinning motors within a single window, it provides across-run robustness to evaluate datasets generated at different steady-state hover speeds.
- **Relative Ratios:** In a standard two-blade configuration, the aerodynamic Blade Pass Frequency (BPF) naturally occurs at 2Ω. The algorithm evaluates the relative $2\Omega/1\Omega$ amplitude ratio and compares it against the baseline healthy BPF signature.
- **Axial-to-Radial Fusion:** A 1Ω radial anomaly alone is insufficient for classification, as it also appears during simple mass unbalance. The discriminator for a bent shaft is the simultaneous presence of the phenomenological F_2 component in the Z-axis alongside the 1Ω radial rise.

This multi-feature separation is defined by the following signature matrix:

Fault Class	**Radial (X/Y)**	**Axial (Z)**	**Distinguishing Feature**
Healthy Baseline	–	2Ω (BPF)	Baseline mechanical and aerodynamic noise.
Mass Imbalance	1Ω Peak	–	Purely radial centrifugal forcing ($F_z = 0$).
Shaft Misalignment	1Ω Peak	2Ω Peak	High $2\Omega/1\Omega$ ratio with Z-axis coupling.
Bearing Defect	–	–	High-frequency envelope sidebands (f_i, f_o, f_b).